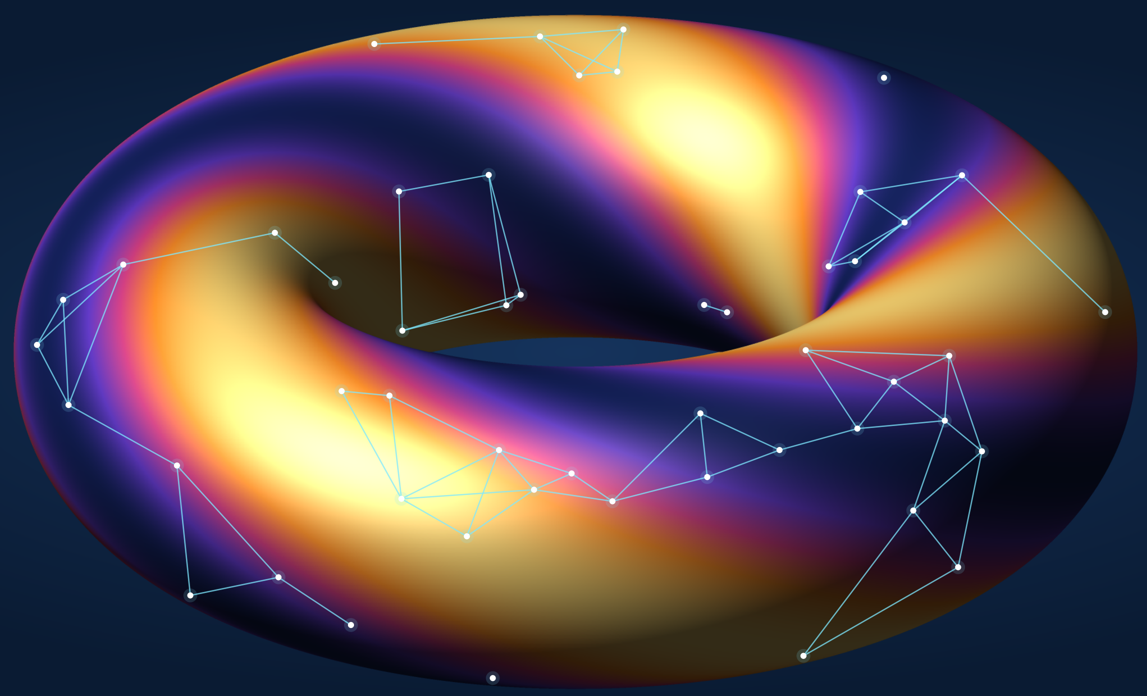
AI FOR GEOMETRY

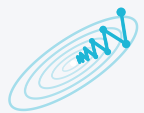
MM845 — Tópicos de Geometria III

30 h contact time · 2 credits · 13 lectures + 13 tutorials · individual project



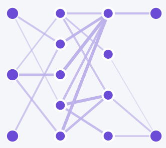
SYLLABUS

Course delivered in English · no prior AI or programming experience required



I Computational & Mathematical Foundations

Scientific programming in Python, version control and reproducibility for experiments in geometry. Learning as a variational problem: optimisation in high dimensions, functional approximation, spectral analysis.



II Supervised Learning & Function Approximation

Linear and logistic models for geometric properties; gradient methods as discretised flows. Neural networks as function spaces for geometric quantities. Convolutions as symmetry-compatible linear operators.

III Unsupervised & Reinforcement Learning

Attention and transformers acting on structured geometric data. Clustering, spectral analysis and dimensionality reduction on manifolds, graphs and simplicial complexes. Reinforcement learning in combinatorial and geometric settings.

IV Geometry-Aware Machine Learning

Equivariance, Lie group actions, graph neural networks and symmetry-compatible representations. Spectral methods, learning on manifolds, and the geometric foundations of diffusion models.

V Geometric PDEs & Scientific Machine Learning

Physics-informed neural networks for elliptic, nonlinear and time-dependent geometric PDEs. Classical numerical analysis meeting deep learning, for the computational investigation of geometric problems.

COURSE STRUCTURE

Tuesdays & Thursdays 14h–16h, each a paired (lecture, tutorial)

Project plan due — end of week 5

Report due + presentations — final week

L1	L2	L3	L4	L5	L6	L7	L8	L9	L10	L11	L12	L13
T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13

INDIVIDUAL PROJECT
USING AI

FIRST 6.5 WEEKS

13 × 1 h lectures · 13 × 1 h tutorials

FINAL 2 WEEKS

Summative assessment: individual project using AI in Geometry. Assessed in 2 parts:
(1) Project plan (2 pages); (2) Project report (5 pages) & Presentation (10 mins).

TAUGHT BY

Henrique N. Sá Earp
Lecturer

Edward Hirst
Lecturing Assistant

Tomás S. R. Silva
Teaching Assistant

Instituto de Matemática, Estatística e Computação Científica · UNICAMP · Campinas, Brazil

hqsaearp@unicamp.br ehirst@unicamp.br tomas@ime.unicamp.br

Recommended prerequisites: MA719 Álgebra Linear Avançada · MA720 Análise no $\mathbb{R}^n$